

AFFINE CHANNELS, MIXED-ORDER SECTIONS, AND ONE UPWARD RETURN IN THE ERDŐS–STRAUS SOURCE

THE CLANKERS

ABSTRACT. An explicit subgroup-chain digit section gives a positive integer lower bound for the complete original square-zero divisor coefficient. It uses nonunit steps, retains cyclic carries, and is attained on hard-prime packets missed by the unit-width criterion. On the exact common-trace locus, the E and M fine coordinates differ by a proved affine shift. Their two-channel operator is injective with an alternating inverse and an orbit-parity cokernel; a partial radix chain reduces universal occupancy to an exact quotient-support cover. An original factor exchange then gives a one-shell upward return at the previously terminal trace at $p = 67369$. All results retain their actual source conditions; no universal ES existence assertion is made.

Original source. For prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, set $R = 4a - p$ and retain $u \mid a^2$. Writing $g = \gcd(a, u)$, $h = g^2/u$, $r = u/g$, $s = a/g$, the original ordered maps are

$$E : (a, h\kappa, phr\kappa), \quad \kappa = (pr + s)/R, \quad M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R.$$

Their full gates are $R \mid 4u + 1$ and $R \mid p + 4u$. These are the classical Type-I/II coordinates [1, §2]. Integral rational-tail trace is the weaker condition $R \mid G^2$, with the same respective G . Lopez's Type A/B classification and discussion of $p \equiv 1 \pmod{24}$ provide additional structural context [3]; they are not used as proofs of the new statements below. Let $K = \prod_{\ell^e \parallel R} \ell^{\lceil e/2 \rceil}$ and $D = R/K$. Then $D \mid K$, so the ideal $K\mathbb{Z}/R$ is square-zero. With channel exponent $\eta = 1$ for E and 0 for M, a trace has the unique fine coordinate

$$p^\eta u/a = -1 + Kc \pmod{R}, \quad c \in \mathbb{Z}/D.$$

Its full gate is $c = 0$, and its tail denominator is $D/\gcd(D, c)$.

For each prime $q_i^{e_i} \parallel a$, partition the original centred interval $[-e_i, e_i]$ modulo $o_i = \text{ord}_K(q_i)$ into $\beta_i = b_i + o_i t_i$, $0 \leq t_i < N_i$. On every trace part,

$$c = c_0 - \sum_i \lambda_i t_i, \quad \lambda_i = (q_i^{o_i} - 1)/K \pmod{D}.$$

Hence its exact integer full count is C_{c_0} in

$$C = \prod_i \sum_{t=0}^{N_i-1} [t\lambda_i] \in \mathbb{Z}[\mathbb{Z}/D]. \quad (1)$$

Every original word occurs once. The exact gcd factorization and ordering are reconstructed after any positive coefficient is selected.

Theorem 1 (Exact affine channel operator). *For a fixed original divisor u , put $G_E = 4u + 1$ and $G_M = p + 4u$. The same u is a trace in both channels exactly when*

$$K \mid G_E \quad \text{and} \quad K \mid p - 1.$$

If $K \nmid p - 1$, the two trace sets are disjoint; if $K \mid p - 1$, they coincide. On the common locus, put $\omega = (p - 1)/K$ and retain its class $\bar{\omega}$ modulo D . The exact coordinates are

$$c_E \equiv G_E/K, \quad c_M \equiv p^{-1}G_M/K, \quad c_E = c_M - \bar{\omega} \pmod{D}. \quad (2)$$

Thus the labelled M and E full counts in a common trace part with M base coordinate c_0 are C_{c_0} and $C_{c_0 - \bar{\omega}}$.

Define $(T_{\bar{\omega}}f)_x = f_{x - \bar{\omega}}$ and $J = (I + T_{\bar{\omega}})C$. If $d = \gcd(D, \omega)$ and $L = D/d$, then L is odd,

$$2C_x = \sum_{j=0}^{L-1} (-1)^j J_{x - j\bar{\omega}}, \quad (3)$$

and the sequence

$$0 \longrightarrow \mathbb{Z}^{\mathbb{Z}/D} \xrightarrow{I + T_{\bar{\omega}}} \mathbb{Z}^{\mathbb{Z}/D} \xrightarrow{\pi} (\mathbb{Z}/2)^d \longrightarrow 0 \quad (4)$$

is exact, where π records the coordinate-sum parity on every translation orbit. For nonnegative C , all labelled two-channel counts are positive exactly when

$$\text{supp}(C) \cup (\text{supp}(C) + \bar{\omega}) = \mathbb{Z}/D. \quad (5)$$

This forces the sharp necessary bound $|\text{supp}(C)| \geq (D+d)/2$; condition (5), rather than its cardinality consequence, is exact.

Proof. The trace tests are $K \mid G_E$ and $K \mid G_M$, while $G_M - G_E = p - 1$, proving the locus statement. Since $4a \equiv p \pmod{R}$,

$$pu a^{-1} + 1 \equiv G_E, \quad u, a^{-1} + 1 \equiv p^{-1}G_M \pmod{R}.$$

Dividing the retained multiples of K by K in $K\mathbb{Z}/R \cong \mathbb{Z}/D$ gives (2). Equivalently, multiplying $-1 + Kc_M$ by $p = 1 + K\omega$ gives $-1 + K(c_M - \omega) + K^2\omega c_M$; the final term is zero modulo R because $D \mid K$. This proves the two coefficient indices without changing any bounded exponent interval.

On each $\bar{\omega}$ -orbit, $T^L = I$ and $(I + T)(I - T + \cdots + T^{L-1}) = 2I$, proving (3) and injectivity. An image has even sum on every orbit. Conversely, these parity conditions make every numerator in (3) even, giving an integral inverse; one coordinate vector on each orbit proves that π is onto. Finally $J_x > 0$ exactly when C_x or $C_{x-\bar{\omega}}$ is positive. The missing positions form an independent set in each odd cycle, whose size is at most $(L-1)/2$. This proves (5) and the sharp bound. $\square$

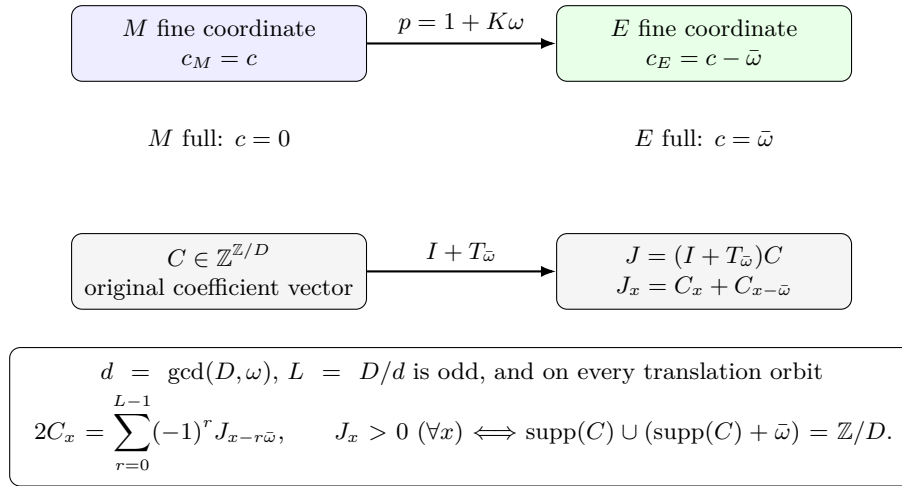


FIGURE 1. Exact common-trace channel shift and odd-orbit inverse.

Theorem 2 (Mixed-order bounded section). Choose distinct indices $i_1, \dots, i_k$, and let $g_0 = D$, $g_j = \gcd(g_{j-1}, \lambda_{i_j})$, $m_j = g_{j-1}/g_j$. Suppose $g_0 > \cdots > g_k = 1$ and $N_{i_j} \geq m_j$. Then

$$C_x \geq \left(\prod_{i \notin I} N_i \right) \prod_j \lfloor N_{i_j}/m_j \rfloor > 0 \quad (x \in \mathbb{Z}/D). \quad (6)$$

The D bounded digits $0 \leq t_{i_j} < m_j$ give a bijection with $\mathbb{Z}/D$. The empty chain for $D = 1$ is included. If all selected $N_{i_j} = m_j$, equality holds in (6).

Proof. The image of λ_{i_j} generates $g_j\mathbb{Z}/D$ modulo $g_{j-1}\mathbb{Z}/D$, a cyclic quotient of order m_j . For a residue $c_k = x$, invert backwards:

$$t_{i_j} = (c_j/g_j)(\lambda_{i_j}/g_j)^{-1} \pmod{m_j}, \quad 0 \leq t_{i_j} < m_j, \quad c_{j-1} = c_j - \lambda_{i_j}t_{i_j} \pmod{D}. \quad (7)$$

The remaining residue is divisible by g_{j-1} , ending in zero. Each quotient forces its unique digit. Partition each selected interval into full blocks of size m_j . Every choice of blocks and of other coordinates translates the required residue and gives one distinct original word by (7), proving (6). $\square$

Theorem 3 (Partial-chain quotient criterion). Allow the chain to stop at $g = g_k > 1$. Put

$$B = \prod_j \lfloor N_{i_j}/m_j \rfloor, \quad Q_y = \#\{(t_i)_{i \notin I} : \sum_{i \notin I} \lambda_i t_i \equiv y \pmod{g}\}.$$

Then

$$C_x = BQ_{x \bmod g} + E_x, \quad E_x \geq 0, \quad (8)$$

with every $E_x = 0$ if $m_j \mid N_{i_j}$ for all selected coordinates. For the common channels of Theorem 1, with $\delta = \omega \pmod{g}$,

$$C_x + C_{x-\bar{\omega}} \geq B(Q_{x \bmod g} + Q_{(x \bmod g) - \delta}). \quad (9)$$

It follows that the complete-block subsource isolated by this partial chain forces all two-channel targets exactly when

$$\text{supp}(Q) + \{0, \delta\} = \mathbb{Z}/g. \quad (10)$$

If $d_g = \gcd(g, \delta)$, condition (10) requires the sharp lower bound $|\text{supp}(Q)| \geq (g + d_g)/2$. If the gcd of g with all residual steps is greater than one, (10) is impossible.

Proof. Retain every complete radix block in each selected interval. A fixed tuple of block offsets lies in $g\mathbb{Z}/D$, and the within-block decoder is a bijection onto that subgroup. Each residual tuple whose sum is x modulo g therefore gives one word over every one of the B block tuples. Words meeting a terminal interval form $E_x \geq 0$, proving (8) and its equality case. Formula (9) is (8) at the two exact channel targets. Formula (10) follows because all coefficients are nonnegative. Its support bound is the odd-cycle argument from Theorem 1. If all residual sums lie in a proper subgroup $b\mathbb{Z}/g$, the complete-block support and one translate meet at most two of its $b \geq 3$ cosets, so they cannot cover $\mathbb{Z}/g$. Terminal intervals remain in the original coefficient vector and may add further words. $\square$

The digit set is not given the wrong product-group law: in $\mathbb{Z}/9$ with steps $(6, 1)$ and radices $(3, 3)$, the actual carry is $(0, 2) \star (0, 1) = (2, 0)$, not $(0, 0)$. The exact multiplicity is also retained. In the equality case every bin has $\mu = \prod_{i \notin I} N_i$ original words; the free-integer coefficient kernel has rank $D(\mu - 1)$. After choosing one reference word in each residue fibre, its differences from the other words in that fibre form a basis. The decoder inverts the augmented residue-and-unused- digit map; the residue projection alone has μ -point fibres.

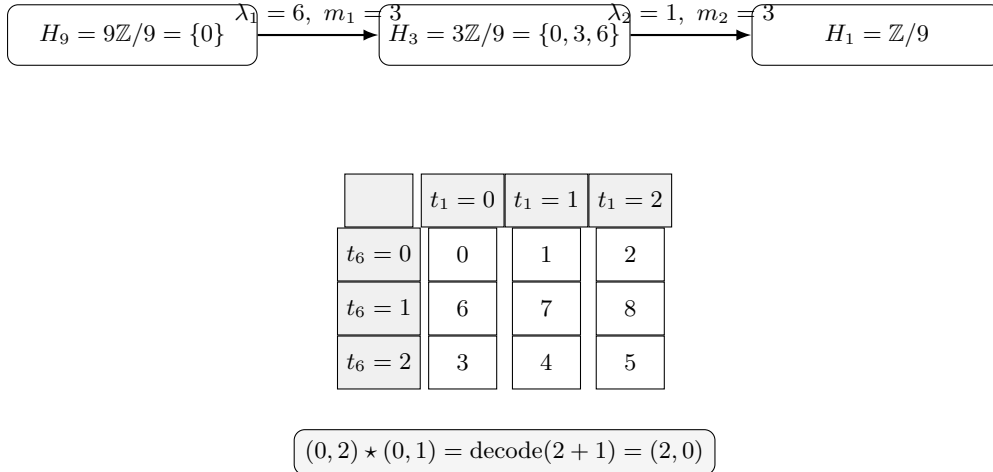


FIGURE 2. The $9 \rightarrow 3 \rightarrow 1$ subgroup chain, exact residue section and nontrivial carry.

All possible chain lower bounds can be optimized on a divisor DAG: use $g \rightarrow \gcd(g, \lambda_i) < g$ when N_i is at least the quotient order, with edge weight $\lfloor N_i/m \rfloor / N_i$. Multiply the maximal path product by $\prod_i N_i$. An effective coordinate cannot recur because every later gcd already divides its step. Thus at most $n\tau(D)$ candidate edges suffice. Failure of this test is not absence of the target coefficient. Arithmetic beyond unit directions. For

$$R = 243, \quad a = H \cdot 53 \cdot 163 \cdot 271, \quad p = 4a - R, \quad H \equiv 29 \pmod{210},$$

retain the actual M packet with centred exponents $\{-1, 1\}, [-1, 1], [-1, 1]$ at the three displayed primes, all other centred exponents zero. It has

$$\mathcal{C} = (1 + X^5)(1 + X^6 + X^{12})(1 + X + X^2) = 2 \sum_{i=0}^8 X^i \pmod{X^9 - 1}.$$

The chain $9 \rightarrow 3 \rightarrow 1$ has radices $3, 3$. The previous total unit width is only three, below eight. One original M return is $(h, r, s, \lambda) = (H, 44173, 53, 182)$.

For

$$R = 675, \quad a = H \cdot 89 \cdot 271^2 \cdot 1801, \quad p = 4a - R, \quad H \equiv 101 \pmod{210},$$

use centred exponents $\{-1, 1\}, [-2, 2], [-1, 1]$. Now

$$C = (1 + X^{11}) \left(\sum_{j=0}^4 X^{6j} \right) (1 + X^{10} + X^{20}) = 2 \sum_{i=0}^{14} X^i \pmod{X^{15} - 1}.$$

Both selected steps 6, 10 are nonunits; their radices are 5, 3. The only unit width is one. An original M return is $(H, 1801, 6536249, 9686)$. Each displayed p -family is a reduced linear progression modulo $840A$, where A is the displayed fixed product. Modulo its prime factors, $p = -R$ is a unit; modulo 840, $p = 1$. Dirichlet supplies infinitely many prime members [2]. No arbitrary-prime coverage is inferred.

The last capacity bound is arithmetically necessary for uniform completion. For an odd defect D , fix a strict chain $g_0 = D$, $g_j = \gcd(g_{j-1}, \lambda_j)$, $g_k = 1$, and put $m_j = g_{j-1}/g_j$, $m = m_k$, $R = 3D^2$, $K = 3D$. Realize earlier steps by primes $q_i = 1 + K\lambda_i \pmod{R}$ with exponents $(m_i - 1)/2$. For the last step choose $q_k = -1 + Kc \pmod{R}$, $2c = -\lambda_k \pmod{D}$, with exponent $m - 1$. Since $K^2 = 3R$,

$$q_k^2 \equiv 1 - 2Kc \pmod{R}, \quad (q_k^2 - 1)/K \equiv \lambda_k \pmod{D}.$$

For the earlier centred exponents and the odd last exponents,

$$z \equiv -1 + K \left(c\beta_k - \sum_{i < k} \lambda_i \beta_i \right) \pmod{R}.$$

The earlier exact-radix intervals biject onto $H_m = m\mathbb{Z}/D$. The last $m - 1$ exponents, multiplied by the unit $c \pmod{m}$, give every nonzero quotient class and no zero. The exact fine count is zero on H_m and one elsewhere, with stabilizer exactly H_m . The CRT conditions $p = 1 \pmod{840}$, $p = 4A - R \pmod{4A^2}$ are compatible and reduced, proving infinitely many hard-prime original packets with that defect. Other source words are retained; this does not furnish or claim a conjectural counterexample.

The separate finite realization is

$$D = 9, \quad (q_1, q_2) = (163, 107), \quad H = 523, \quad a = 976015801, \quad p = 3904062961,$$

with coefficient vector $(0, 1, 1, 0, 1, 1, 0, 1, 1)$ and stabilizer $\{0, 3, 6\}$. Its exact-exponent progression is $p = 3904062961 + 2925429291933960t$. The displayed $R = 243$ shell is empty, but the same prime has the full E witness $(a_0, R_0, u_0) = (976015741, 3, 89)$, so this is not an ES counterexample.

Theorem 4 (A labelled upward factor exchange). *For an original shell and chosen $s \mid a$, put $C = R + s$ and $M = p + C = 4a + s$. All cells with the same labelled invariant are*

$$\begin{aligned} s' \mid M, \quad M/s' &\equiv 1 \pmod{4}, \quad 0 < C - s' < p, \\ a' &= (M - s')/4, \quad R' = C - s', \quad h' = (M/s' - 1)/4. \end{aligned} \tag{11}$$

Such a cell gives an original unit-ray E state precisely when $R' \mid M$. Then $u' = h'$, $r' = 1$, $\kappa' = M/R' - 1$, and its ordered denominators are $(a', h's'\kappa', ph'\kappa')$.

Proof. The congruence is precisely the integral factorization $a' = h's'$ and implies $R' \equiv 3 \pmod{4}$. The range gives the original first-half domain. Since $s' \mid a' < p$, one has $\gcd(R', s') = \gcd(4h's' - p, s') = \gcd(p, s') = 1$. Thus the original E gate $R' \mid 4h' + 1$ is equivalent to $R' \mid M = (4h' + 1)s'$. Equivalently, $p + s' = M - R'$, so $R' \mid p + s'$ is exactly $R' \mid M$. Retaining s, s' inverts (11), with $a' - a = (s - s')/4$. $\square$

At $p = 67369$, the proper trace $(a, R, u, h, r, s) = (16849, 27, 4067, 83, 7, 29)$ has $C = 56$, $M = 67425$. Taking $s' = 25$, $R' = 31$ gives $a' = 16850$, $h' = 674$, $\kappa' = 2174$ and

$$\frac{4}{67369} = \frac{1}{16850} + \frac{1}{36631900} + \frac{1}{98714178844}.$$

Every earlier shell is fully exhausted in the certificate and has no integer gate. The return must increase a at this input. More generally, set $h = 83 + 4650k$, $p = 812h - 27 = 67369 + 3775800k$. On every prime member, the same proper trace $(203h, 27, 49h)$ returns to

$$a' = 203h + 1, \quad R' = 31, \quad h' = (203h + 1)/25, \quad r' = 1, \quad s' = 25, \quad \kappa' = (p + 25)/31.$$

All divisions are exact and the prime progression is reduced. The previous word/ray deletions and factor-sum return fail on that marked source. No assertion that every old shell is wholly empty along this progression is needed.

At $p = 97, a = 40, u = 2$ in M, the invariant instead has $C = 83, M = 180$. Its entire cell list is $(a', R', s') = (36, 47, 36), (40, 63, 20), (44, 79, 4)$, each with both full gates empty. Thus a fixed invariant fibre need not be occupied. Universal existence still requires an argument linking original sources across shells; neither theorem assumes that missing first source.

$$p = 67369, \quad C = 56, \quad M = p + C = 67425$$

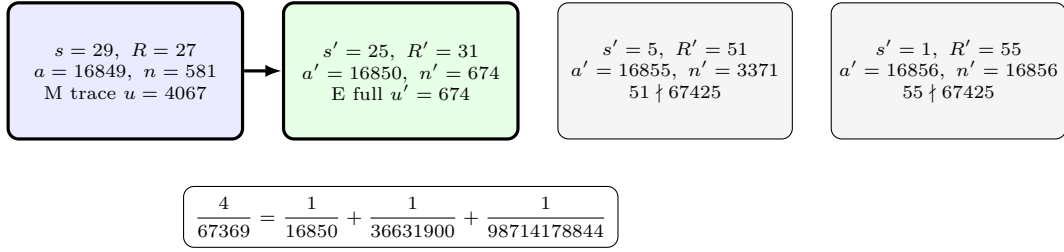


FIGURE 3. All four cells in the $p = 67369, C = 56$ fibre and the exact selected upward arrow.

Complete incoming source, rather than an unmarked inverse. For a supplied primitive target with $r' = 1$, recover $C = R' + s'$ and $M = p + C$. Enumerate every cell in (11). For each cell enumerate $r \mid n$ with $\gcd(r, s) = 1$ and set

$$h = n/r, \quad u = nr, \quad a = ns.$$

The E and M source tests are respectively

$$R \mid (pr + s)^2, \quad R \mid (r + s)^2.$$

These finite tests are necessary and sufficient for all canonical trace inputs whose retained s produces this C . The factorization of the old shell gives $n = hr$ and recovers exactly the displayed data; conversely the exact gcd factorization of $a = ns, u = nr$ gives (h, r, s) because $\gcd(r, s) = 1$. An M word already records its ordered tails. On ordered rational triples the exact E swap is $\tau_E(a, Y, Z) = (a, Z, Y)$ with $\tau_E^2 = \text{id}$; it does not generally preserve the original E word box and is not inserted into the canonical incoming fibre. The forward arrow records its chosen target; forgetting that choice gives a relation whenever several terminal cells pass. The free-integer kernel of the arrow projection has a basis formed by choosing one reference arrow in each target fibre and subtracting it from every other arrow in that fibre. It is not a quotient that identifies the old and new factor supports.

For the target at $p = 67369$, the complete cells have (R, s) equal to

$$(27, 29), (31, 25), (51, 5), (55, 1).$$

Only $(31, 25)$ is an exterior terminal cell. Its canonical incoming trace fibre contains precisely the original M word 4067 at $a = 16849$ and the full E word 674 at $a = 16850$. Thus the source-changing arrow crosses the obstruction; its complete fibre does not assert that a corresponding arrow exists at every prime. At $p = 97$ the three cells given above have no full original gate even when all their words, not just the unit-ray words, are tested.

Evidence. The accompanying full source gives every inverse fibre and finite bound. The standard-library checks compare the complete original boxes through the declared prime bound and exhaust 161887 additive polynomials, with a separate subsequence-based optimum check and independent primitive-state enumeration. The files distinguish observed runs, finite example searches and written universal proofs. No human review, Lean build or priority determination is claimed.

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